

KHARTOUM INTL, Sudan

WMO: 627210

Lat: 15.589N

Lon: 32.553E

Elev: 386

StdP: 96.78

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.8	16.1	-13.1	1.3	32.6	-11.4	1.5	29.9	10.0	24.9	9.3	23.8	5.4	0	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.5	43.2	19.5	42.7	19.5	41.8	19.4	25.6	32.6	25.0	32.5	24.5	32.3	4.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.8	29.1	23.0	18.7	28.5	22.2	17.7	28.8	81.5	32.9	78.2	31.9	76.1	31.9	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.9	8.1	DB	12.3	45.0	1.6	0.6	11.2	45.5	10.2	45.8	9.4	46.2	8.2	46.6	(4)
(5)			WB	4.9	26.3	0.9	1.3	4.2	27.2	3.7	28.0	3.2	28.7	2.5	29.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.8	24.2	26.9	29.7	33.0	35.2	35.2	33.0	31.2	32.8	33.0	29.3	25.4	
	DBStd	4.32	3.32	3.14	2.88	2.41	1.87	1.74	2.33	2.20	2.13	1.85	2.55	2.85	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	7579	439	473	610	690	782	757	714	658	685	712	579	478	
	CDD18.3	4538	182	240	352	440	524	507	456	400	435	454	329	220	
	CDH23.3	68581	1899	2932	5006	7182	8973	8771	7249	5866	6795	7190	4398	2320	
	CDH26.7	44446	863	1587	3077	4928	6504	6376	4830	3514	4451	4761	2496	1059	
Wind	WSAvg	4.5	5.0	5.1	4.9	4.5	4.1	4.2	5.0	4.8	3.9	3.6	4.4	4.8	
Precipitation	PrecAvg	120	0	0	0	0	3	3	37	56	19	3	0	0	
	PrecMax	447	0	0	0	0	25	15	185	235	88	16	0	0	
	PrecMin	4	0	0	0	0	0	0	0	0	0	0	0	0	
	PrecStd	92	0	0	0	0	6	4	42	65	19	4	0	0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	40.0	42.8	44.1	44.8	44.2	42.2	40.5	41.9	41.2	39.1	36.8	
		MCWB	17.4	17.9	18.9	19.8	20.0	19.2	20.1	22.6	20.9	20.4	19.5	18.8	
	2%	DB	35.1	37.8	40.9	42.8	43.7	43.1	41.1	39.0	40.5	40.2	37.9	34.9	
		MCWB	17.7	17.8	17.8	19.6	19.3	19.6	19.9	22.8	21.1	20.2	19.1	18.2	
	5%	DB	33.2	36.1	39.0	41.2	42.8	42.1	40.1	37.9	39.2	39.8	36.9	33.7	
		MCWB	17.4	17.2	17.3	18.6	19.2	19.7	20.6	22.8	21.6	20.2	18.9	18.0	
	10%	DB	31.2	34.2	37.1	40.1	41.8	41.1	39.0	36.7	38.2	38.8	35.8	32.1	
		MCWB	16.5	16.6	16.4	18.1	19.0	19.6	21.2	23.0	21.7	20.3	18.4	17.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	20.1	20.2	23.6	24.0	24.7	25.6	26.5	26.7	24.3	21.9	20.9
			MCDB	32.2	35.8	39.8	34.1	36.5	32.6	32.6	32.8	33.1	33.2	35.9	32.2
2%		WB	18.8	18.9	19.0	21.7	22.9	23.5	24.7	25.5	25.8	23.5	20.6	19.6	
		MCDB	32.4	35.1	38.2	37.3	36.1	34.8	32.5	32.2	33.0	33.2	35.4	32.5	
5%		WB	17.9	18.0	18.1	20.8	22.1	22.7	24.1	24.9	25.1	22.8	19.8	18.8	
		MCDB	32.1	34.5	36.9	38.6	36.0	35.0	32.3	32.2	32.6	33.0	35.0	31.9	
10%		WB	16.9	17.1	17.1	19.5	21.5	22.1	23.5	24.5	24.5	22.2	18.9	17.9	
		MCDB	30.9	33.0	35.7	38.6	35.9	35.0	32.2	31.9	32.2	33.0	34.3	30.8	
Mean Daily Temperature Range	5% DB	MDBR	13.1	13.7	14.1	13.8	12.2	11.5	10.6	9.5	10.5	11.2	12.0	12.4	
		MCDBR	14.3	14.8	15.2	14.5	13.5	12.4	11.6	10.9	11.5	11.8	12.4	13.0	
		MCWBR	6.3	6.4	6.3	5.9	5.4	4.6	4.0	3.2	4.1	4.7	5.4	5.7	
	5% WB	MCDBR	13.8	14.1	14.2	13.3	10.9	11.3	10.3	9.7	10.2	10.9	12.3	12.6	
		MCWBR	6.6	6.8	6.4	5.7	4.0	4.3	3.4	3.3	3.9	4.0	6.1	6.3	
Clear-Sky Solar Irradiance	taub	0.423	0.465	0.514	0.557	0.584	0.647	0.695	0.708	0.651	0.562	0.459	0.424		
	taud	2.061	1.939	1.809	1.730	1.711	1.614	1.586	1.575	1.652	1.800	1.982	2.051		
	Ebn at Noon	861	842	811	775	743	690	660	658	694	746	817	845		
	Edh at Noon	160	189	221	240	241	262	270	276	255	215	172	158		
All-Sky Solar Radiation	RadAvg	5.84	6.54	7.24	7.59	7.35	7.22	6.72	6.40	6.49	6.29	5.94	5.57		
	RadStd	0.09	0.09	0.15	0.26	0.17	0.17	0.29	0.23	0.20	0.11	0.07	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page